BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 237 · Testing and program quality

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why should Largest() require a nonempty input array?

Q2. What is a regression?

Q3. What does a class method do?

Q4. A date validator passes all unit tests but booking fails when writing to the database. What test is needed?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

unit testing — Test one module/function in isolation.

integration testing — Check interaction of components.

system testing — Test the entire application as a system.

acceptance testing — Check fitness for user/business requirements.

regression testing — Repeat tests to catch defects after changes.

New learning

Test individual units — Unit tests isolate one function or class method from the rest of the system.

Test connections and behaviour — Integration tests interactions between modules; system tests the complete application.

Confirm intended outcome — Acceptance testing checks whether the delivered application meets stakeholder requirements. Automated regression tests rerun after changes.

Worked example

A checkout computes the right total but fails to save orders. Which test is most relevant?

1. Predict first — TotalCalculator already gives correct values; the failure appears when OrderService interacts with the database.

2. Model and explain — Prioritise an integration test of saving, then a system-level end-to-end checkout test.

We Do · Practise with guidance

A search function works alone but the website shows the wrong results. What should be tested?

```
Search() returns expected IDs.
Website displays mismatched titles.
```

Predict first

The isolated search passes, so check the connection that maps returned IDs to displayed titles.

Model and explain

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

An integration test can expose the mapping mismatch, while a system test checks the complete user-visible result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Match each: individual ValidDate function, database interaction, full website journey, client sign-off to the appropriate test level.

Question 2

A login test passes until a new password rule is introduced, after which registration fails. Explain a sensible debugging and testing workflow.

Question 3

Why is 100% execution of code lines not equivalent to proving an application correct? Suggest three further forms of evidence.

Exit Ticket · Answer before checking

Passing unit tests alone proves that the complete application works for its users.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It reads a[0] as its starting value.

Q2. Previously working functionality fails after a change.

Q3. Defines behaviour for objects of that class.

Q4. An integration test of the database interaction.

You Do pages · Independent questions

Q1. [4] Unit [1], integration [1], system [1], acceptance [1].

Q2. [6] Reproduce failure [1]; inspect changed rule and expected behaviour [1]; step through registration [1]; correct fault [1]; rerun registration unit/integration tests [1]; rerun previous login tests for regression [1].

Q3. [6] Executing every line may not test all branches or input combinations [2]; use boundary and erroneous test cases [1], integration tests [1], realistic user acceptance scenarios [1] and regression tests after changes [1].

Exit Ticket · Model answer

False. Interactions, full-system behaviour and user requirements require further testing.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding